Invasive Species Threats Assessment

The Restoration of Cisco in Lake Erie

Summary Report **June 2024**

A. SPECIES OF CONCERN

The following species have been identified as having an impact on cisco Coregonus artedi from the GLANSIS database and have a broad distribution within the Lake Erie basin.

Rainbow smelt (*Osmerus mordax*)

- **Competition**
  - High probability of habitat overlap, niche overlap in spring-summer diet is high (Rosinski et al. 2020).
  - Historically correlated with cisco decline, including its congeners (Christie 1974; COSEWIC 2007).
- **Predation**
  - Predation by rainbow smelt accounts for 15–52% and 37–100% of the mortality of larval ciscoes in Thunder and Black bays, Ontario (Myers et al. 2009).
  - Declines, local extirpations, and limitations to recovery of cisco populations have also been attributed to rainbow smelt predation on larval fish (Stockwell et al. 2009).

Spiny waterflea (*Bythotrephes longimanus*)

- **Competition**
  - Experimental studies in Harp Lake, Canada suggest that *Bythotrephes longimanus* likely competes with other vertebrate predators for zooplankton, such as Coregonus artedi (Dumitru et al. 2001).
- **Disease/Parasite/Toxicity**
  - Significant increases in the trophic position of zooplankton (reduced herbivorous cladoceran biomass and increased omnivorous/predatory copepod biomass) and planktivorous fish such as the lake herring (*Coregonus artedi*) with the introduction of *B. longimanus* have the potential to lead to substantial contaminant biomagnification in consumers (Rennie et al. 2011). No reported cases of contamination at this time.
- **Beneficial**
  - Spiny waterflea is a food source for *Coregonus artedi* (Branstrator and Lehman 1996).

Sea lamprey (*Petromyzon marinus*)

- **Disease/Parasites/Toxicity**
  - Evidence of impact by predation on *Coregonus* spp., sea lamprey predation led to the extinction of the deepwater cisco (*C. johannae*) and the decline of the blackfin cisco (*C. nigripinnis*), both endemic to the Great Lakes (Miller et al. 1989).
  - Petromyzon marinus is expected to parasitize *Coregonus artedi* in areas where they co-occur, similar to what has been seen with lake trout in Lake Ontario (Schneider et al. 1996).

Alewife (*Alosa pseudoharengus*)

- **Competition**
  - Researchers speculated that cisco native to Lake Michigan, shifted habitat and diet as a result of competition with alewife (Crowder 1984).
- **Genetic**
  - Crowder (1984) speculated that a cisco native to Lake Michigan, the bloater (*Coregonus hoyi*) evolved fewer and shorter gill rakers as a result of competition with alewife.

Round goby (*Neogobius melanostomus*)

- **Predation**
  - Recent evidence presented at IAGLR 2024 suggests that *Coregonus artedi* larvae may avoid habitat occupied by round goby. Cisco larvae exhibited tighter group cohesion and frequently avoided round goby in experimental trials (Hoyer et al. 2024).
- **Beneficial**
  - Round goby has been incorporated into the diet of *Coregonus artedi* in Lake Ontario (Gatch et al. 2021).

B. SPATIAL ANALYSIS

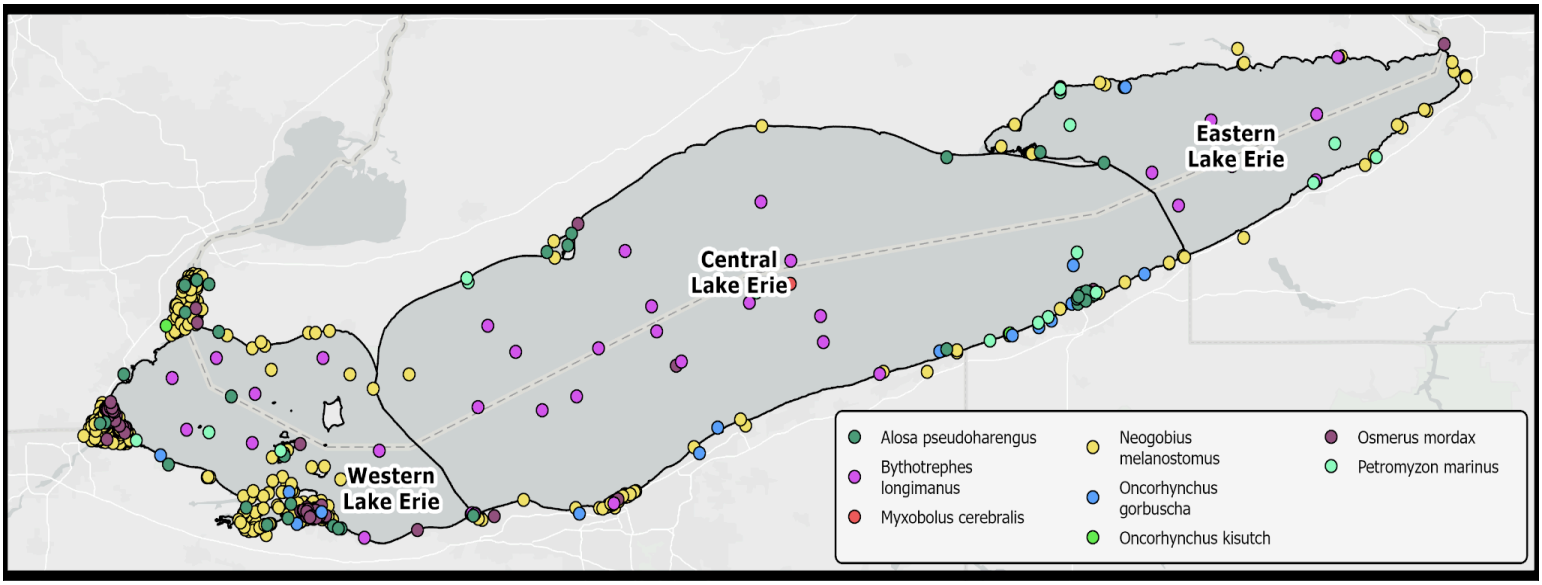


Figure (Above): Distribution data for all AIS of interest (Source: GLANSIS).

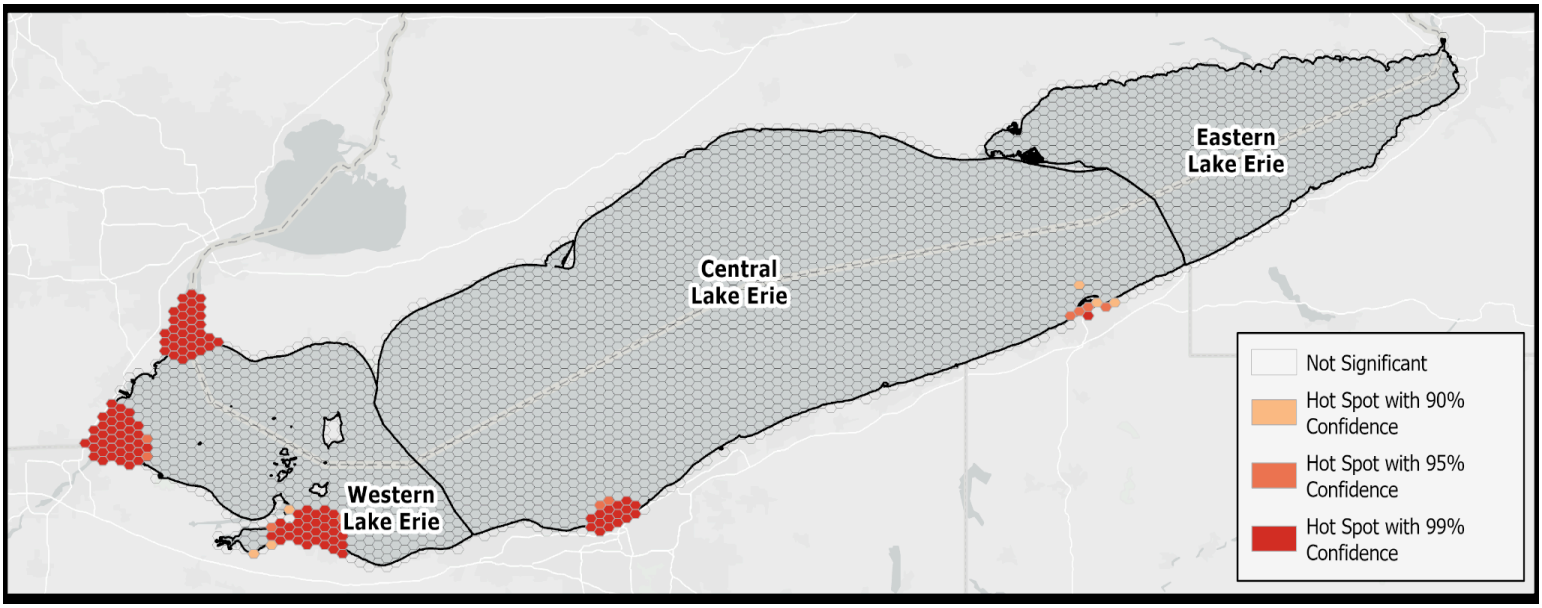


Figure (Above): Optimized Hot-spot Analysis of AIS Distribution Data.

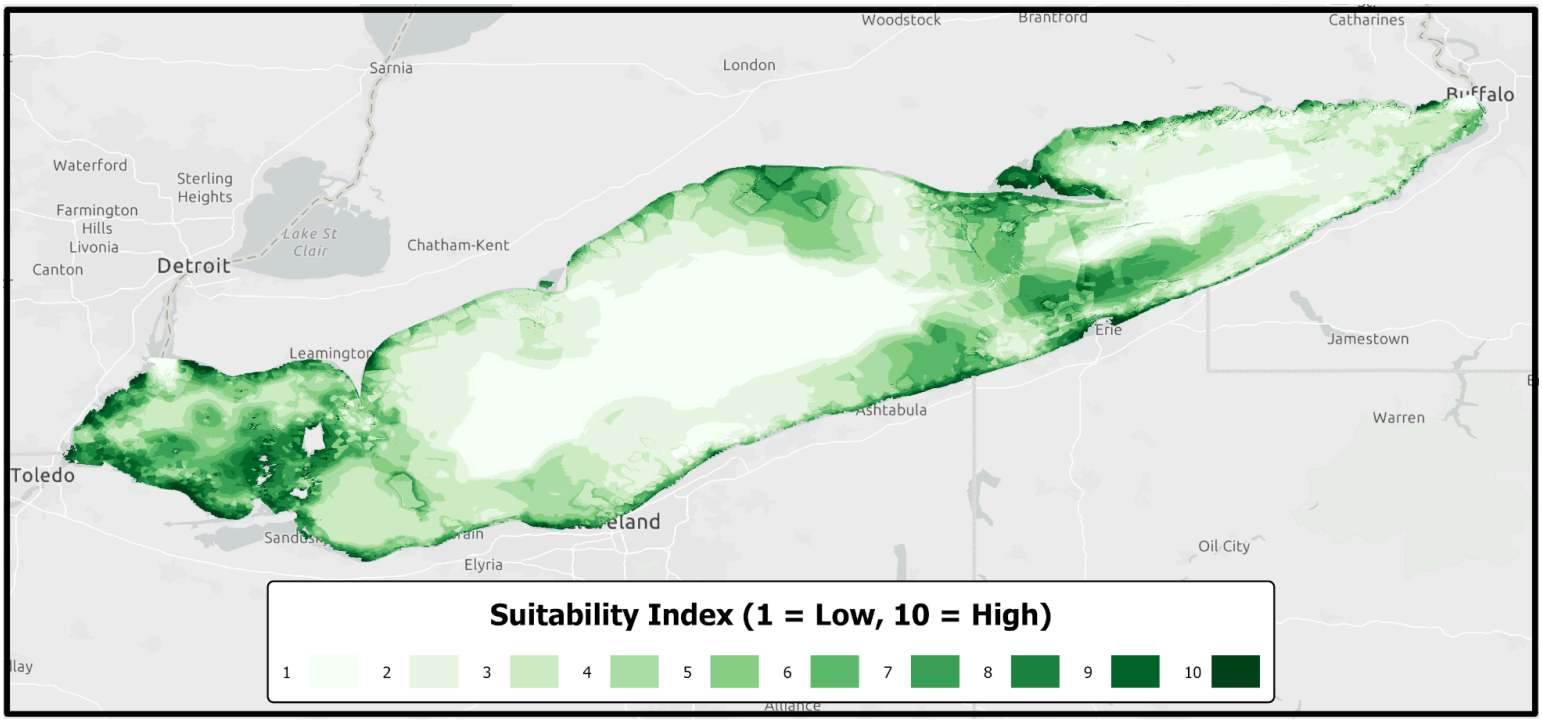


Figure (Above): Suitability raster for *Coregonus artedi* reproductive habitat, sourced from a 5-fold cross validation MaxEnt model with historical substrate, historical coefficient of variation of ice duration, predicted reef locations, fetch, circulation, and distance to delta wetlands as variables. Credit: Katelyn King, U of M SEAS.

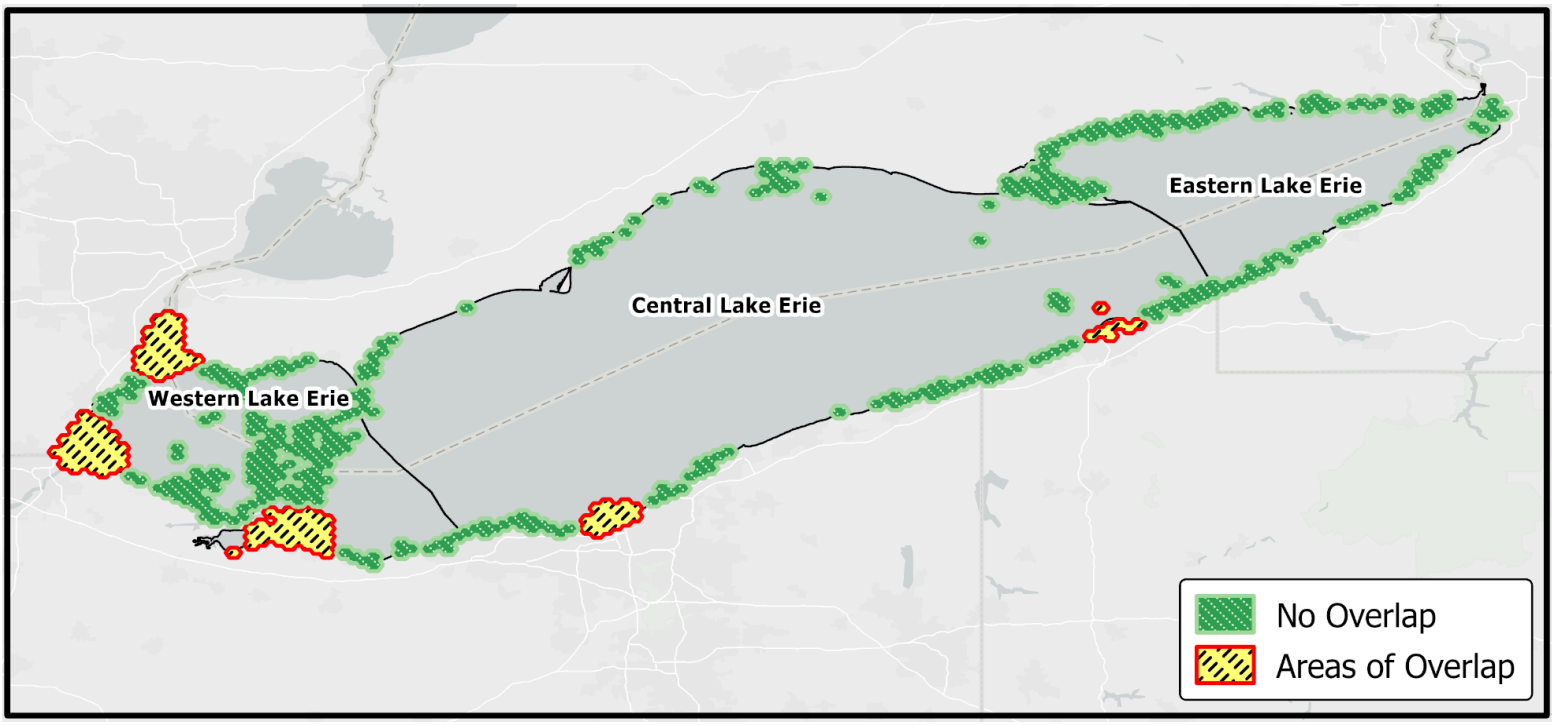


Figure (Above): 'Areas of Overlap' = areas of high suitability (≥ 7.5 on suitability index) AND hot spot with 99% confidence
'No Overlap' = areas of high suitability (≥ 7.5 on suitability index) AND not significant.

C. REFERENCES

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